1. (a) Test statistic is $\chi^2 = \frac{(n-1)S^2}{100}$. Decision rule: reject H_0 if $\chi^2 \le \chi^2_{1-\alpha}(n-1)$. $H_0: \sigma^2 = 100$ is null hyp; $H_1: \sigma^2 < 100$ is alternative/research hyp.

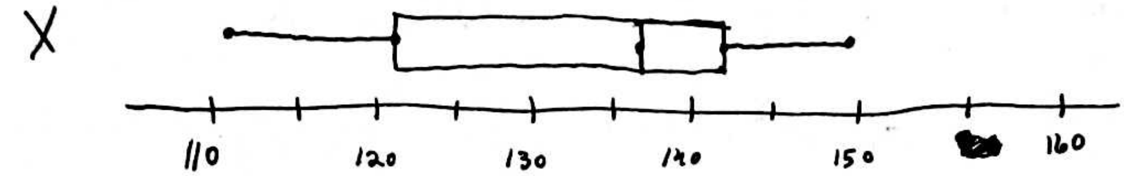
(b) $\chi^2 = \frac{22 \cdot (32.52)}{100} = 7.154$, $\chi^2_{0.975}(22) = 10.98 \Rightarrow$ reject H_0

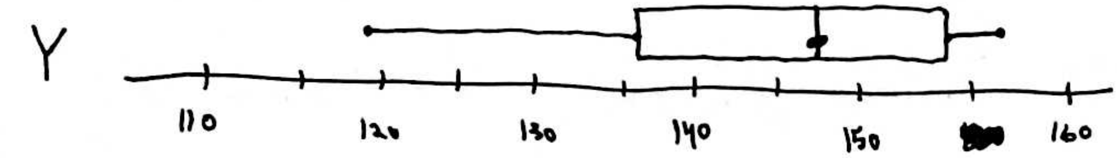
(c) Reject H_0 if $\chi^2 \le \chi^2_{1-\alpha/2}(n-1)$ or $\chi^2 \ge \chi^2_{\alpha/2}(n-1)$ (see Table 8.3-1)

8.2.6 (a) Since both are normal with same variance, we use equations (8.2-1) & (8.2-2): statistic is $T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{1/n_x + 1/n_y}}$, where $S_p = \sqrt{\frac{(n_x - 1)S_X^2 + (n_Y - 1)S_Y^2}{n_x + n_Y - 2}}$. If H_0 is true, then T has a t-distribution with $r = n_x + n_Y - 2$ deg. of freedom. For a 2-sided alternative hyp. with $\alpha = 0.05$, we reject H_0 when $|t| \ge t_{0.025}(n_x + n_Y - 2)$ (Table 8.2-1).

(b) $s_x^2 = 198.3$, $s_Y^2 = 150.4 \longrightarrow S_p = 13.2042$; $\bar{x} = 131.5$, $\bar{y} = 144.2$
$\Rightarrow t = -2.151$. $t_{0.025}(18) = 2.101 < |t|$. So, we reject H_0; i.e., we conclude that the two studs require different amounts of force to remove.

(c) Since $t_{0.025}(18) = 2.101$ and $t_{0.01}(18) = 2.552$, the p-value is between 0.05 and 0.02.





The box plots appear to confirm that X and Y have different means.

<u>8.2.10</u> Since the variances are unknown/not assumed to be equal and the samples are small, we use $W = \dfrac{\bar{X} - \bar{Y}}{\sqrt{S_X^2/n_X + S_Y^2/n_Y}}$ (equation 7.2-3), which has approximately a t-dist. with $r = \dfrac{\left(S_X^2/n_X + S_Y^2/n_Y\right)^2}{\frac{1}{n_X-1}\left(S_X^2/n_X\right)^2 + \frac{1}{n_Y-1}\left(S_Y^2/n_Y\right)^2}$ (eq. 7.2-1). Here X is the normal air population, Y the enriched air. We calculate $s_X^2 = 0.9143$, $\bar{x} = 4.163$, $s_Y^2 = 2.591$, $\bar{y} = 5.105 \Rightarrow w = -1.489$, $r = \lfloor 10.3 \rfloor = 10$. Since $t_{0.1}(10) = 1.372$ and $t_{0.05}(10) = 1.812$, the p-value is between 0.1 and 0.05 (since we are using a one-sided alternative hypothesis). So, we can be 90% but not 95% confident that the enriched air increases growth.

<u>8.4.2</u> (a) For example, reject H_0 if $X \le 1$. Then $C = \{x : x \le 1\}$. $(x = 0 \text{ or } x = 1)$

(b) $\cdot\ \alpha = \mathbb{P}(X \le 1; p = 3/5)$. Since $X|(p = 3/5)$ is $Bin(4, 3/5)$, $\alpha = \binom{4}{0}(3/5)^0(2/5)^4 + \binom{4}{1}(3/5)(2/5)^3 = 0.0208$.

$\cdot\ \beta = \mathbb{P}(X > 1; p = 2/5) = 1 - \mathbb{P}(X \le 1; p = 2/5) \overset{\text{(b/c } X|(p=2/5) \text{ is } Bin(4,2/5))}{=} 1 - \left[\binom{4}{0}(2/5)^0(3/5)^4 + \binom{4}{1}(2/5)^1(3/5)^3\right] = 0.5248$

8.4.10 (a) Using table 8.4-1, test statistic is $Z = \frac{Y/n - 0.65}{\sqrt{0.65(0.35)/n}}$, & critical region is $Z \geq z_\alpha = 1.96$.

(b) $Z = \frac{414/600 - 0.65}{\sqrt{0.65(0.35)/600}} = 2.054$.

(c) p-value $= P(W \geq 2.054)$ where $W \sim N(0,1)$. From the table, this is between 0.0202 and 0.0197. So we can be about 98% confident in rejecting H_0.

(d) Lower bound is $\underbrace{\hat{p}}_{=414/600} - 1.645\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.6589$ (compare top of page 386); interval is $[0.6589, 1]$.

8.5.3 (a) There are $y = 8$ observations below $m_0 = 5.9$. From table II ($n = 25, p = 0.5$), we get the p-value 0.0539. Since this is slightly above $\alpha = 0.05$, we do not reject H_0.

(c) Since the t-distribution is symmetric, its mean equals its variance; thus, we test $H_0: \mu = 5.9$ against $H_1: \mu > 5.9$. The test statistic is $T = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}$, $\mu_0 = 5.9$, and we reject H_0 if $t \geq t_\alpha(n-1)$. We get $t = 2.608$. Since $t_{0.05}(24) = 1.711 < t$, the t-test also rejects H_0. p-value is between 0.005 and 0.01 (from table VI).

(b) The values of $|x_i - 5.9|$ are, in the same order as the data,

0.275	-21
0.235	-19
0.203	-17
0.063	-6
0.037	-5
0.03	-4
0.022	-3
0.016	-2
0.008	1
0.067	7
0.119	8
0.12	9
0.129	10
0.132	11
0.137	12
0.145	13
0.149	14
0.15	15
0.179	16
0.216	18
0.259	20
0.286	22
0.299	23
0.407	24
0.487	25

Therefore W = sum of right column = 171. The approximate p-value is $P(W \geq 171) \approx$

$$P\left(\frac{W}{\sqrt{(25)(26)(51)/6}} \geq \frac{171-1}{\sqrt{(25)(26)(51)/6}}\right) \approx P(Z \geq 2.287),$$

where we used $171 - 1 = 170$ because of the "continuity correction" (see 5.7, also pages 396, 397). The last probability is between 0.0113 and 0.011 (from table Va), so we reject H_0 at the $\alpha = 0.05$ significance level.